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Effects of microchannel confinement on acoustic vaporisation of ultrasound phase change contrast agents

Shengtao Lin¹, Ge Zhang, Chee Hau Leow and Meng-Xing Tang^{1,2}

Department of Bioengineering, Imperial College London, London, United Kingdom

E-mail: mengxing.tang@imperial.ac.uk

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Abstract

The sub-micron phase change contrast agent (PCCA) composed of a perfluorocarbon liquid core can be activated into gaseous state and form stable echogenic microbubbles for contrast-enhanced ultrasound imaging. It has shown great promise in imaging microvasculature, tumour microenvironment, and cancer cells. Although PCCAs have been extensively studied for different diagnostic and therapeutic applications, the effect of biologically geometrical confinement on the acoustic vaporisation of PCCAs is still not clear. We have investigated the difference in PCCA-produced ultrasound contrast enhancement after acoustic activation with and without a microvessel confinement on a microchannel phantom. The experimental results indicated more than one-order of magnitude less acoustic vaporisation in a microchannel than that in a free environment taking into account the attenuation effect of the vessel on the microbubble scattering. This may provide an improved understanding in the applications of PCCAs *in vivo*.

Keywords: ultrasound imaging, phase change contrast agent, acoustic vaporisation, microvascular, microbubble, image quantification

(Some figures may appear in colour only in the online journal)

1. Introduction

Ultrasound imaging with microbubble (MB) contrast agents have offered substantial benefits and extended applications for clinical practise e.g. cardiology and radiology (Cosgrove and Harvey 2009, Stride and Edirisinghe 2009). There have been significant advances in

¹ Current address: Department of Bioengineering, Imperial College London, London, SW7 2AZ, United Kingdom.

² Author to whom any correspondence should be addressed.

developing contrast-enhanced ultrasound imaging (Yeh *et al* 2015, Stanziola *et al* 2016), and MB-mediated therapeutics (Yeh 2012, Cosgrove 2015). However, MBs are mainly limited to intravascular contrast tracers because of their micron size being larger than inter-endothelial gaps (Cosgrove 2006). Hence, a smaller ultrasound contrast agent is essential to explore the extravascular space. Phase-change contrast agent (PCCA), which has the sub-micron initial size, has opened up new opportunities for both diagnosis and image-guided therapy (Rapoport 2012, Sheeran and Dayton 2012, Williams *et al* 2013). PCCAs can be potentially used for cancer extravascular imaging via the enhanced permeability and retention effect (Torchilin 2011). The defective tumour vasculature potentially allows extravasation of PCCAs through leaky endothelial layers while the reduced lymphatic drainage of tumours gives rise to prolonged retention time of extravasated PCCAs in the tumour tissue (Prabhakar *et al* 2013). The phase-transition (liquid phase to gaseous state) of perfluorocarbon in the core of a PCCA can be triggered by directed external acoustic energy with a desired temporary and spatial specificity, which the activation phenomenon is named acoustic vaporisation (of PCCAs). The formation of echogenic MBs via acoustic vaporisation provides ultrasound echo signals *in situ*. Much of previous studies on PCCA have focused on characterising acoustic vaporisation (Reznik *et al* 2013, Williams *et al* 2013, Sheeran *et al* 2014, Li *et al* 2015, Novell *et al* 2016), contrast-enhanced molecular imaging (Rapoport *et al* 2011, Sheeran *et al* 2013b), drug/gene delivery (Burgess and Porter 2015, Chen *et al* 2015, Gao *et al* 2015, Kasoji *et al* 2015, Marshalek *et al* 2016), ablation/occlusion therapy (Kripfgans *et al* 2002, Moyer *et al* 2015), focused ultrasound therapy (Schad and Hynynen 2010, Kang and Yeh 2011, Chen *et al* 2013, Liu *et al* 2016), dual-modality imaging (Lin *et al* 2017), and tissue engineering (Moncion *et al* 2016). Most recent research has shown that PCCAs can be created from commercially available ultrasound contrast agents (Sheeran *et al* 2017) for medical ultrasound imaging purpose, which may facilitate the potential clinical translation in the future.

A number of studies have demonstrated that MB dynamics can be affected by the surrounding medium geometry. Thomas *et al* (2013) investigated MB oscillations in capillary tube phantoms (diameters: 160 μm and 25 μm) and revealed that a reduction of MB oscillating amplitude can be up to 50% when MBs are constrained within a microvessel. A similar experimental study from Caskey *et al* (2006) also found that the expansion of MB diameter under ultrasound within a small diameter (12 μm and 25 μm) microvessel tubes was less than that in a 195 μm one. By observing MB dynamics under high-speed camera, Garbin *et al* (2007) demonstrated that the oscillation amplitude was suppressed by at least 50% in the vicinity of the wall. It also has been demonstrated on a branched microvascular phantom (Jamalian *et al* 2016) that there was a 40% decrease in the averaged MB population scattering signal magnitude when the vessel-channel size decreases from 200 μm to 50 μm (Lin *et al* 2016b). In addition, numerical simulation/modelling also have provided insight into the understanding MB dynamics relating to MB-vessel interaction (Qin and Ferrara 2006, Hosseinkhah *et al* 2013), the effect of boundary proximity (Helfield *et al* 2014, Alexander and Ayache 2015).

To our knowledge, the effect of geometrical confinement *in vivo*, such as microvasculature, tissue interstitium, tumour microenvironment, and the cytoplasm in the cells, on the efficiency of acoustic vaporisation of PCCAs is still not clear. A large proportion of blood volume circulates in the microvascular network, especially the cancerous tumour, where resides different types of small vessels with diameters ranging from a few hundreds of micrometers to several micrometers (i.e. capillaries). Besides the microvasculature, PCCAs can also extravasate into tissue interstitium or intracellular delivery into cancer cell by receptor-mediated endocytosis (Marshalek *et al* 2016) prior to vaporisation. Therefore, for *in vivo* applications, an understanding of PCCA imaging with geometrical confinement is needed for e.g. interpreting image findings, optimising ultrasound parameters for acoustic vaporisation, PCCA dosage,

and informing the development of PCCA-specific theranostic tools etc. Recent efforts on *in vivo* studies of PCCAs revealed that the vaporisation efficiency may be reduced; the same pulse parameters used for *in vitro* phantom experiment were not capable of generating comparable level of contrast for *in vivo* case (Puett *et al* 2014, Sheeran *et al* 2015, Sheeran *et al* 2016). For instance, (Sheeran *et al* 2016) demonstrated imaging and activating PCCAs both *in vitro* and *in vivo* in the 20–40 MHz frequency range using pre-clinical ultrasound platform. The acoustic vaporisation of PCCAs *in vitro* can be initiated at an output power as low as 3% (~1.3 MPa) of output power whereas that in tumour and kidney can be achieved with a power level up to 100% (~7.1 MPa) but no less than 40% (~6.0 MPa). Such a large difference cannot be entirely explained by tissue attenuation and aberration.

In this study, we demonstrated and quantified, for the first time, the effects of microvessel confinement on acoustic vaporisation of PCCA populations *in vitro*. The acoustic vaporisation of PCCAs in an open environment (a water tank) was compared with that in microchannel confinement (a 200 μm -cellulose tube). Our quantification results have revealed that acoustic vaporisation of PCCAs in the 200 μm -cellulose tube is significantly different from that in the open environment with the post-vaporisation acoustic signals being suppressed by more than ten-fold.

2. Method

2.1. PCCAs preparation

The PCCAs were prepared using ‘MB condensation’ approach described in detailed previously (Matsunaga *et al* 2012, Li *et al* 2015). Briefly, the sub-micron PCCAs were obtained via condensation of precursor lipid-coated perfluorocarbon (decafluorobutane)-filled MBs. 1,2-dipalmitoyl-sn-glycero-3-phosphocholine (DPPC), 1,2-dipalmitoyl-sn-glycero-3-phosphoethanolamine-N-[methoxy(polyethylene glycol)-2000] (16:0 PEG2000 PE) in a 9:1 molar ratio was mixed and dissolved in propylene glycol, glycerol, and phosphate-buffered saline (PBS) (15/5/8, v/v/v). All lipids used in this study were purchased from Avanti Polar Lipids, Inc., USA. 1 ml of this excipient solution was transferred to a sealed 2 ml glass vial and its headspace was exchanged with decafluorobutane gas (Fluoromed, USA). Then, the precursor MBs were produced by standard mechanical agitation (Vialmix, USA). Finally, the PCCA emulsion was generated via the condensation of precursor MBs with pressurisation in a salt-ice bath ($-7\text{ }^{\circ}\text{C}$ to $-10\text{ }^{\circ}\text{C}$). The precursor MBs were sized and counted using the protocol described in (Sennoga *et al* 2010). The size distribution of PCCAs was obtained using dynamic light scattering (DLS, Malvern Nano ZetaSizer, UK). One ml of a 10% diluted PCCA solution, i.e. 100 μl of the stock PCCA in 900 μl deionised and distilled water, was placed in a sizing cuvette before measurement. Each measurement was produced by three repeated records, and the final size result was an average of three measures for three vials. The concentration of PCCA emulsion was determined using the method in Li *et al* (2015).

2.2. Experiment setup

To simulate the geometrical confinement, an optically and acoustically transparent 200 μm -cellulose tube (Cuprophane, RC55 8/200 Membrana GmbH, Germany) was utilised in this study as shown in figure 1. The cellulosic membrane was made of a natural linear polysaccharide material (Ong *et al* 2013) with a wall thickness of 8 μm . Given the acoustic attenuation of such material (Cuccaro *et al* 2015) and the wall thickness, its attenuation is assumed negligible. This is also consistent with other MB-related studies (e.g. Owen *et al* 2012, 2015) showing the optical

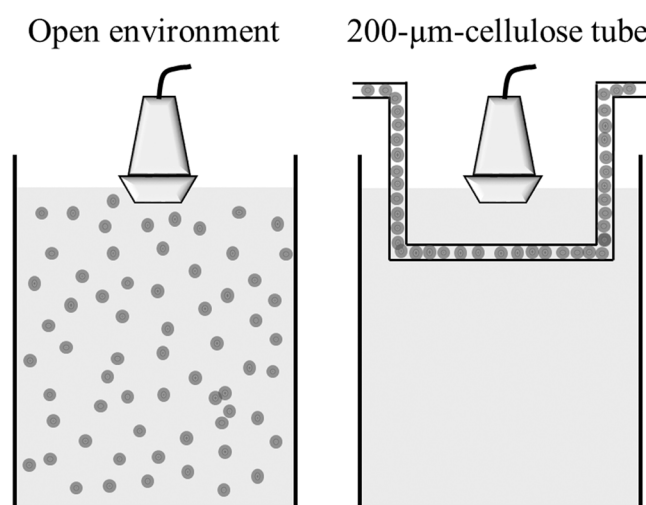


Figure 1. Schematic representation of the experimental set-up (not to scale). The same concentration of PCCA solution was introduced into open environment and 200 μm -cellulose tube at physiological temperature 37 $^{\circ}\text{C}$.

Table 1. ‘Imaging-activation-imaging’ sequence and ultrasound parameters for PCCA imaging.

State	Pulse type	Transmit frequency (MHz)	Pulse length (cycle)	PNP (MPa)	MI
Imaging (before)	Plane-wave	4.5	1	0.15	0.07
Activation	Focus	8	5	4.81	1.70
Imaging (after)	Plane-wave	4.5	1	0.15	0.07

and acoustically transparency of the tube, and our own experimental measurement of the tube with a hydrophone where no significant acoustic attenuation was found. For comparison, water in an acrylic-walled 2 l tank was assumed to be an ‘open environment’ with no geometrical confinement. Before the microtube experiment the diluted PCCA solution was prepared to achieve a dilution factor of 0.1% relative to the stock emulsion concentration (0.2 ml stock emulsion into 200 ml water), which reached 5.5×10^6 particles ml^{-1} (refers to the concentration details in section 3.1). The same dilution factor and concentration were also applied to the open environment PCCA solution by scaling the amount for both stock PCCA emulsion and water in the tank (slowly infusing 2 ml stock emulsion into 2 l water), and the solution was gently stirred before each experimental run to avoid sedimentation and uniformity. For the microtube experiment, the diluted PCCA solution was gently shaken in the syringe before they were slowly infused into the tube followed by clamping both ends of the tube to stop the flow inside, and waiting for 15 s prior to initiating the whole pulse sequence. Each experiment repetition was performed with refilling the fresh PCCA solution into the tube. The water used in the experiment was the deionised water that reached gas equilibrium (Mulvana *et al* 2012) prior to experiment. Acoustic absorbers were used to line the water tank to reduce ultrasound reflections. This experiment setup was also employed to determine microbubble factor (MBF) detailed in the section 2.7.

2.3. PCCA imaging sequence

A custom designed ‘Imaging-Activation-Imaging’ sequence (Puett *et al* 2014, Li *et al* 2015, Lin *et al* 2016a) was applied on a research platform (Verasonics Vantage 128, Kirkland, USA) equipped with a L12-5 50 mm linear array probe (ATL, USA). The central 128 elements (the maximum available number of connector channels) were used for the entire sequence. The ultrasound parameters were detailed in table 1. In ‘Activation’ sequence, electronically steering focus pulses (8 MHz, 5-cycle sinusoidal waves with mechanical index (MI) of 1.70, f -number = 0.68, pulse-repetition-frequency = 14.3 kHz, total duration of exposure is 8.8 ms) were transmitted to vaporise PCCAs and produce MB-based ultrasound contrast enhancement in the microcellulose tube and open environment. The depth of axial focus (i.e. spatial and temporal peak-negative-pressure, PNP) of the activation-pulse was predetermined as 17 mm, at which the microcellulose tube was positioned during experiment. To measure the MB-specific imaging contrast, pulse-inversion (PI) technique (Stanziola *et al* 2016) transmitting plane-waves of 15-angle spatial compounding (angle range: -18° to 18°) was utilised in both ‘Before’ and ‘After’ imaging sequences with an image acquisition frequency of 100 Hz. Each plane wave was a single-cycle pulse at 4.5 MHz with a MI of 0.07. A pair of ‘Positive’ and ‘Negative’ images was collected for each image so that both PI (harmonic) and B-mode (fundamental) signal were available for data analysis.

2.4. L12-5 Probe calibration

To calibrate the pressure field and estimate the focal volume of activation-pulse, i.e. PCCA activation volume in open environment, the ultrasound beam profile of the activation-pulse was measured using a 0.2 mm needle hydrophone (mentioned above) in the equilibrated water environment (Mulvana *et al* 2012). Firstly, the pressure field in the axial direction was measured by moving the hydrophone using a stepping motor (Velmex, UK) with 0.1 mm interval. Then, the elevational pressure profile was calibrated; the needle hydrophone was initially positioned at the lateral centre of the probe and at the focus depth (i.e. 17 mm) and then it was moved elevationally by the motor with 0.05 mm interval. For the plane-wave, the measure was conducted in an identical fashion.

2.5. Activation volume factor (AVF)

When activating PCCAs dispersed in the open environment, all the PCCAs within the effective focal volume of the activation-pulse can undergo phase-shift and generate MBs. However, for 200 μ m-cellulose tube experiment, this activation volume was different as the PCCAs were confined within the microcellulose tube. While the axial and lateral dimension of this volume on the ultrasound image to be quantified can be the same by choosing the same region of interest (ROI) on the image, the elevational dimension of the volume was different: it was 200 μ m for the tube, and was the effective elevational focus length of the activation pulse for the open environment which was determined by the hydrophone measurement. This difference between tube and open environment experiment can be compensated for according to a previous study showing the acoustic power increases (API) linearly with PCCA concentration (Li *et al* 2015). Therefore an AVF was calculated using:

$$\text{Activation Volume Factor (AVF)} = \frac{V_{\text{open environment}}}{V_{\text{microcellulose tube}}} = \frac{L}{d} \quad (1)$$

where $V_{\text{open environment}}$ and $V_{\text{microcellulose tube}}$ are the estimated activation volume for the open environment and microcellulose tube respectively. L is the full-width-half-maximum (FWHM) of the elevational focus length of activation-pulse that applied to activate the PCCAs. d is the diameter of microcellulose tube i.e. $200\ \mu\text{m}$.

2.6. Acoustic Power Increase (API)

The ultrasound images were produced by beamforming the raw radiofrequency (RF) data using a customised program described in Leow *et al* (2015). The spatial dimension of each image was $40\text{ mm} \times 25\text{ mm}$ (axially $\times$ laterally). The pixel dimensions on each coherent plane-wave compounded image were $50\ \mu\text{m} \times 200\ \mu\text{m}$ (axially $\times$ laterally). The ultrasound contrast enhancement produced by the acoustic vaporisation of PCCAs was measured using API defined as below.

$$\text{Acoustic Power Increase (API)} = (\mu_{\text{after}} - \mu_{\text{before}})^2 \quad (2)$$

where μ_{after} and μ_{before} were the mean image pixel amplitude in ‘Before’ and ‘After’ images respectively.

Since the region, where PCCA-generated contrast in the open environment outside the corresponding tube area, can contribute to the maximum signal magnitude in the ROI on the image, an image processing technique was applied to achieve the consistency in the FWHM of the APIs between the tube and the open environment experiment results, such that an unbiased quantification can be performed between the APIs of tube and open environment. This technique simulated the multiple spatially paralleling $200\ \mu\text{m}$ -cellulose tubes within the image plane, which summed both the differential image of PCCA-produced contrast in the tube at its original position and its shifted copies with positions at integer multiples of $200\ \mu\text{m}$ axially centred at the original tube position.

2.7. Microbubble Factor (MBF)

MBF was measured and calculated to account for the attenuation effect of the tube on MB scattering signals—without compensating this even if the same amount of MBs are vaporised in the tube as that in the open environment, the received acoustic signals from the MBs in the tube will be lower. It already has been shown in a number of previous literatures that the MB oscillation and acoustic response were affected by the presence of microvessel confinement and wall vicinity. To quantify the suppression in MB oscillation within the $200\ \mu\text{m}$ -cellulose tube in this study, the echo signal of MB populations in the open environment and in the tube was compared using the same ultrasound plane-wave imaging settings as imaging PCCAs. In the experiment, the precursor MB solution (prior to manufacturing PCCAs) was utilised at a concentration of 10^5 particles ml^{-1} in the tube, and in water tank measurement the concentration was adjusted such that the same estimated number of bubbles were present within corresponding ultrasound imaging volume (taking into account the dimension difference in the elevational direction). The acoustic attenuation of MB population above the selected ROI for the open environment experiment was taken into account as $0.1\ \text{dB mm}^{-1}$ at $4.5\ \text{MHz}$ matching the result in a previously study (Goertz *et al* 2007). The MBF was calculated using:

$$\text{Microbubble Factor (MBF)} = \frac{\text{API}_{\text{in m.t.}}}{\text{API}_{\text{in o.e.}}} = \left(\frac{\mu_{\text{MB in m.t.}} - \mu_{\text{baseline in m.t.}}}{\mu_{\text{MB in o.e.}} - \mu_{\text{baseline in o.e.}}} \right)^2 \quad (3)$$

where $API_{in\ m.t.}$ and $API_{in\ o.e.}$ were the API of MBs in the microcellulose tube and the open environment respectively. $\mu_{MB\ in\ m.t.}$ and $\mu_{MB\ in\ o.e.}$ are the mean image pixel amplitude in the same ROI (at the tube region) for both microcellulose tube and open environment respectively. $\mu_{baseline\ in\ m.t.}$ and $\mu_{baseline\ in\ o.e.}$ are the mean image pixel amplitude of baseline (water) echo strength in the corresponding ROIs for microcellulose tube and open environment respectively showing the background signals. The size and shape of the ROI were $200\ \mu m \times 11\ mm$ (axially $\times$ laterally), rectangle with four-corner coordinates $(-5.5, 16.9)$, $(5.5, 16.9)$, $(-5.5, 17.1)$, and $(5.5, 17.1)$.

2.8. Corrected API

The APIs for both tube and open environment were corrected to quantify the difference in PCCA-generated contrast between the open environment $API_{o.e.c}$ (equation (4)) and the microcellulose tube $API_{m.t.c}$ (equation (5)).

$$API_{in\ o.e.c} = \frac{API_{o.e.}}{AVF} \quad (4)$$

$$API_{in\ m.t.c} = \frac{API_{m.t.}}{MBF}. \quad (5)$$

The final quantification results were produced by calculating the integration values of the laterally averaged $API_{in\ o.e.c}$ and $API_{in\ m.t.c}$ on each image over the FWHM regions along the axial direction.

2.9. Statistical analysis

The means and standard deviations were calculated based on ten individual experiment repetitions. Each repetition produced an average from ten images. Student's *t*-test was used to compare the statistical difference between groups, with $p < 0.05$ being considered significant difference, $p < 0.01$ being highly significant difference.

3. Results

3.1. PCCA and MB characterisation

The normalised size distribution of MBs (figure 2(a)) shows a mean diameter of $1.22 \pm 0.43\ \mu m$, a median of $1.16\ \mu m$, and total concentration of 6×10^9 particles ml^{-1} . PCCA has an averaged diameter of $243 \pm 54\ nm$ (figure 2(b)), an averaged polydispersity index of 0.33, and the concentration of 5.5×10^9 particles ml^{-1} .

3.2. Activation Volume Factor (AVF)

The one-dimension profile of elevational (figure 3) pressure field for the transmitted 8 MHz 5-cycle focus-pulse. The maximum PNP was found at 17 mm (i.e. the peak of axial profile and the centre of elevational profile) where the tube was positioned during experiment. The elevational dimension of the transmitted beam was determined as the FWHM of the beam profile (i.e. 1.2 mm as shown in figure 3), and hence the AVF ($1.2\ mm/200\ \mu m$) was calculated to be 6 (a.u.).

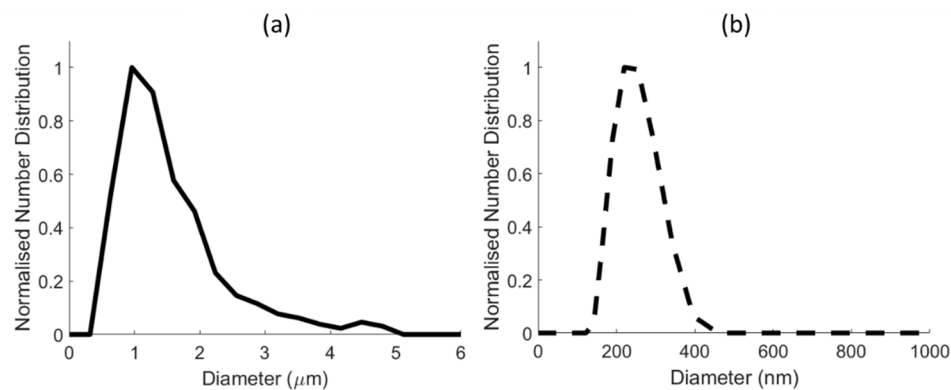


Figure 2. The normalised size distribution of MBs (a) and PCCAs (b). MBs have a mean diameter of $1.22 \pm 0.43 \mu\text{m}$, a median of $1.16 \mu\text{m}$. PCCAs show an averaged diameter of $243 \pm 54 \text{ nm}$ and an averaged polydispersity index of 0.33.

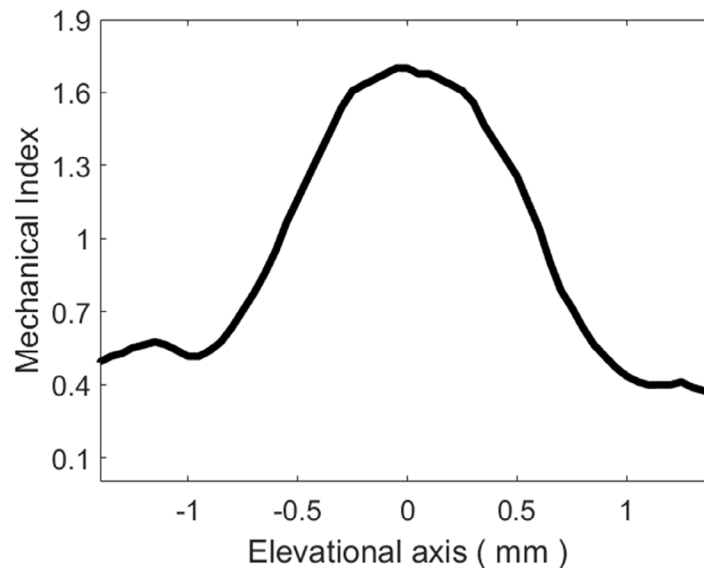


Figure 3. Elevational beam profile of an 8 MHz 5-cycle focus-pulse transmitted from L12-5 probe for activating PCCAs. The calibration result shows a spatial and temporal peak pressure (equivalent to the PNP of a MI 1.7) at 17 mm away from the probe surface and FWHM in elevational direction is 1.2 mm.

3.3. Microbubble Factor (MBF)

The baseline and MB images in the open environment and 200 μm -cellulose tube are shown in figure 4. MBF was calculated to be 0.63 ± 0.09 (analysed using PI images) and 0.59 ± 0.12 (analysed using B-mode images).

3.4. PCCA imaging

Figure 5 show representative PI images of PCCAs before ((a) and (c)) and after ((b) and (d)) acoustic activation in the open environment and the 200 μm -microcellulose tube. In the

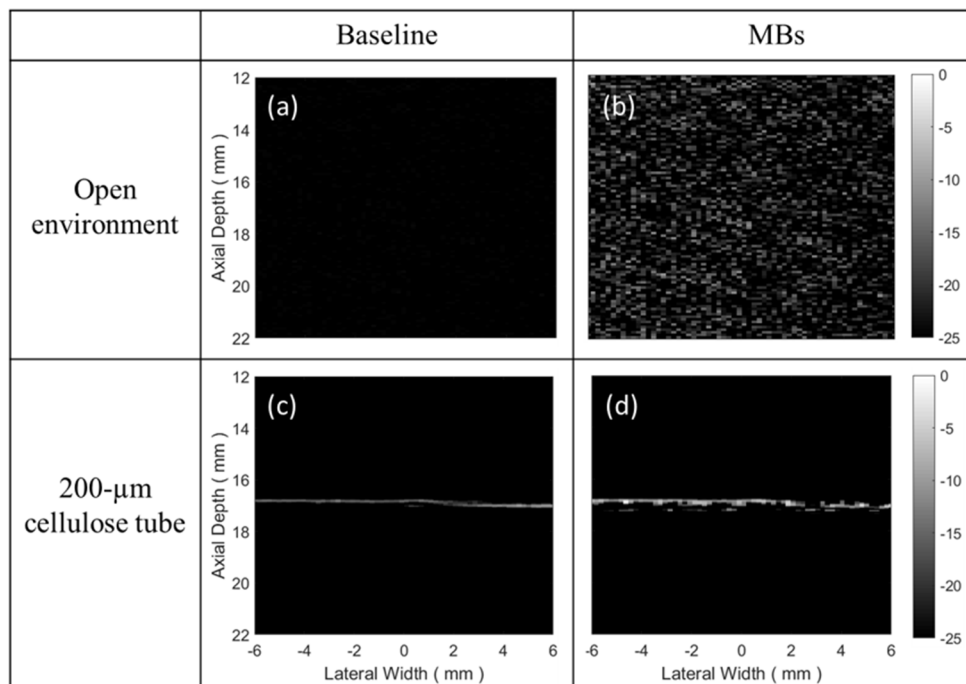


Figure 4. Representative PI imaging MB populations in the open environment (b) and 200 μm -cellulose tube (d) for determining of the suppression factor of MB oscillation in the tube. (a) and (c) show the baseline images of water in the tank and 200 μm -cellulose tube, respectively. The images are presented in a 25 dB scale. The size and shape of the ROI were 200 $\mu\text{m} \times 11\text{ mm}$ (axially $\times$ laterally), rectangle with four-corner coordinates $(-5.5, 16.9)$, $(5.5, 16.9)$, $(-5.5, 17.1)$, and $(5.5, 17.1)$. The MBF was calculated as 0.63 ± 0.09 for Pulse Inversion signals and 0.59 ± 0.12 for B-mode signals.

open environment, MBs were activated mainly around the line-shaped focus region while the PCCAs in the 200 μm -microcellulose tube were vaporised and provided contrast increase within the tube area.

The corrected $\text{API}_{\text{o.e.c}}$ and $\text{API}_{\text{m.t.c}}$ values produced from PI and B-mode images were presented in figure 6. Figure 6(a) revealed that there was highly significant difference ($p = 9 \times 10^{-5}$) between the API produced by PCCAs-induced contrast enhancement in open environment and 200 μm -cellulose tube with an averaged 19.5-fold difference. In figure 6(b), it shows that there was an even larger significantly difference (averaged 27.4-fold) for the API between the open environment and 200 μm -cellulose tube, which suggests the API difference was more dominated by the fundamental frequency imaging component (i.e. 4.5 MHz).

4. Discussion

In summary, we demonstrate the first experimental measurement to understand the effects of microvascular confinement on acoustic vaporisation of PCCAs. A significant decrease in contrast enhancement via acoustic vaporisation of PCCA populations was observed in a micro-channel (200 μm -cellulose tube) when compared with that in an open environment (water tank). The API with the confinement in harmonic (PI) and fundamental (B-mode) signals

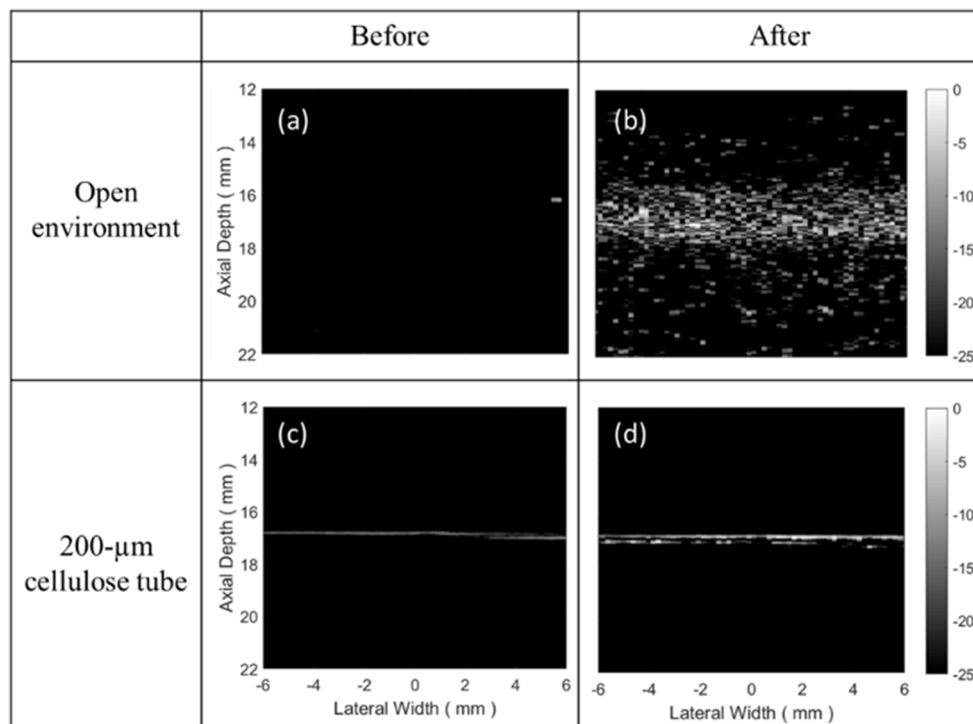


Figure 5. (a) and (b) Representative images before and after activation of PCCAs in the open environment; (c) and (d) images before and after activation of PCCAs in the 200 μ m-cellulose tube. The images were in a 25 dB scale.

were suppressed by an average of 19.5-fold and 27.4-fold respectively. For API quantification results, it is interesting to see the difference in the contrast enhancement between the fundamental B-mode and harmonic PI images (see figure 6). This may indicate that microvascular confinement could alter the size population of newly created MBs from acoustic vaporisation of PCCAs.

There are a few studies investigating acoustic vaporisation process (Doinikov *et al* 2014) and its physical mechanism (Shpak *et al* 2014) using theoretical modelling, however, at present, the effect of spatial constraints on acoustic vaporisation has yet to be fully uncovered and studied. The surface tension and viscosity of a constrained media may affect PCCA dynamics especially in the initial ‘overexpansion phase’ before a vaporised PCCA experiencing damped radial oscillation (Doinikov *et al* 2014). In addition, the energy barrier or probability for homogeneous nucleation of the superheated-state PCCA is likely raised due to the alteration in heat transportation and dissipation by the geometrical constraints (Mountford *et al* 2015). Finally, the presence of microvessel confinement may impact the ‘superharmonic focusing’ (Shpak *et al* 2014) and how it affects the focusing is not clear.

In this study, much efforts were taken to compensate the additional difference between the two phantoms besides the geometry, which mostly includes the elevational focus volume and the suppression on MB scattering signals with vessels. To resolve this, we apply two factors AVF and MBF accordingly. Admittedly, this approach may not be optimal, and may still introduce some errors. For example, it is difficult to exactly determine the difference in activation

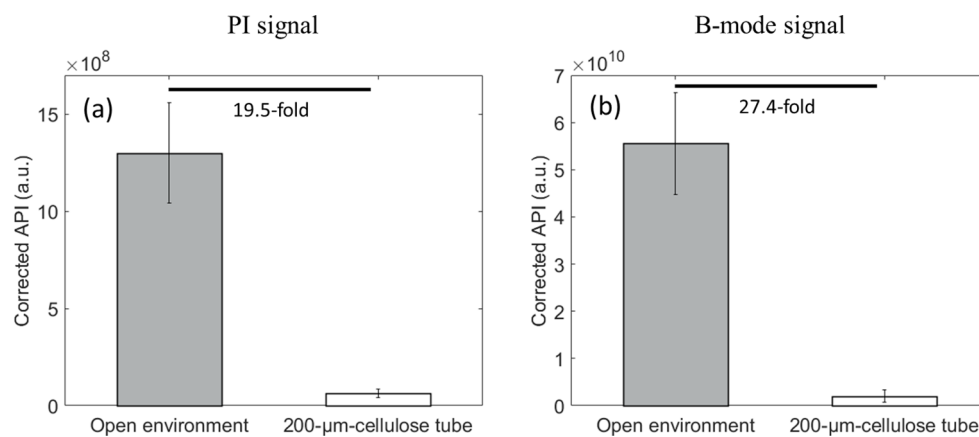


Figure 6. (a) The $API_{o.e.c}$ (grey bar) and $API_{m.t.c}$ (white bar) analysed using PI images. It indicated a highly significant difference ($p = 9 \times 10^{-5}$, an average of 19.5-fold). (b) The $API_{o.e.c}$ and $API_{m.t.c}$ analysed using B-mode images. It shown that an average of 27.4-fold difference in PCCA-produced ultrasound contrast increase after activation between with and without a microvessel confinement. It should be noted that the y-axis were set differently for both figures for the purpose of visualising the significant difference.

volume for the open environment phantom. AVF was estimated using the elevational focal length determined by FWHM of the elevational beam profile of the activation-pulse. However, there was almost no detectable vaporised PCCAs (MBs) at the MI level where FWHM was (i.e. 0.85) based on the evidence shown in figure 7, which suggested the API generated at the activation MIs of 0.7 and 0.9 was below 3% of that produced at a MI of 1.70. This finding was also consistent with the vaporisation threshold of PCCAs described in Sheeran *et al* (2012). So the AVF value can be over-estimated and thus the API for open environment became under-estimated. Therefore, the aforementioned API difference between in the open environment and within 200 μ m-cellulose tube can be also under-estimated.

4.1. Limitations and future directions

The 200 μ m-cellulose tube used in this study can be considered as a rigid constraint surrounding with PCCAs (Thomas *et al* 2013), and the composition of the small tubes may affect the acoustic response of MBs (Butler *et al* 2010). Admittedly, a complex acoustic environment may exist in the cellulose tube (Qin *et al* 2008), which the acoustic pressure may be different from that hydrophone measurement. In fact, the cellulose tube can be a reasonable preliminary approximation for a flexible capillary constrained by thick tissue (Hynynen *et al* 2001). It was also revealed that the MB oscillation shown no difference in amplitude and nonlinearity within the same diameter of tubes either cellulose- or acrylic-type (Caskey *et al* 2006). In this initial *in vitro* evaluation, 200 μ m-cellulose tube was used to simulate a relatively large microvasculature. In future studies, the effects of various microtube diameters should be studied. Additionally, other surrounding biologically structural constrains, such as tissue interstitium, cancerous microenvironment, cell internal, can potentially influence the PCCA imaging in a way that microvasculature does, and these possibilities need to be explored.

The stock ('native') PCCA emulsion was utilised in this study, the size outliers can potentially influence the acoustic vaporisation in some cases when they presented in the ultrasound activation focal volume. The size distribution of PCCA manufactured from condensing

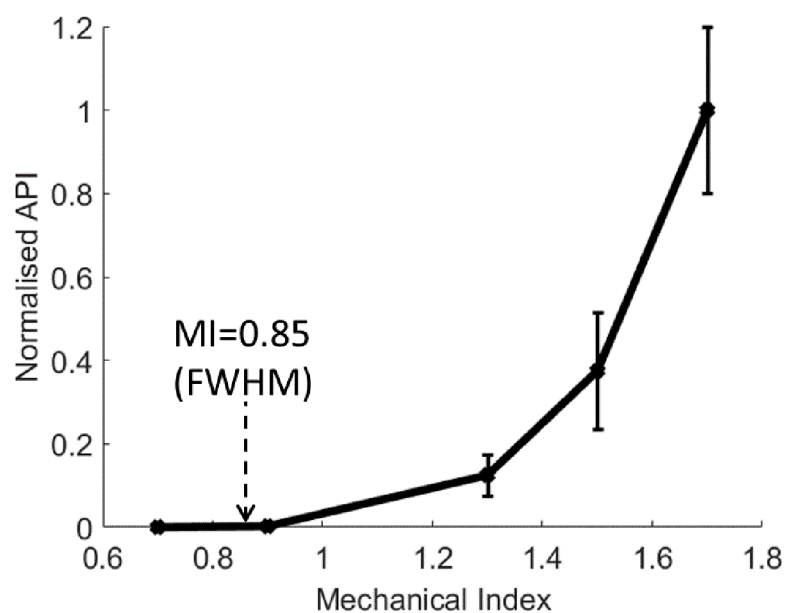


Figure 7. Normalised API as a function of MI of focus-pulse activating PCCAs. It can be interpreted that almost no detectable contrast enhancement at a MI of 0.85 (FWHM) compared with that generated by a MI of 1.70, such that using FWHM to calculate AVF can be conservative. The PCCA API difference between in the open environment and within 200 μm -cellulose tube can be under-estimated using FWHM for AVF.

MBs was polydisperse correlating with the polydispersity of precursor MBs. The size outlier PCCAs can undergo spontaneous vaporisation even without external acoustic energy deposition in a 37 °C environment (Sheeran *et al* 2012). Clinically, this may not be problematic as these outlier MBs do not go to circulatory network before naturally filtered out by the lungs. The ‘decantation’ approach was applied (Sheeran *et al* 2016) to produce ‘size-selected’ PCCAs from the ‘native’ emulsion and the results demonstrated ‘native’ PCCA phantom started generating MBs at PNP of 1.5 MPa while no vaporisation was detected until reaching 3.5 MPa. As a result, the acoustic vaporisation in microvessels could become more difficult when activating a smaller population of PCCAs (‘size-selected’) if also considering the acoustic vaporisation threshold increases as the size of PCCA decrease (Sheeran *et al* 2013a). Future work is required to manipulate PCCA population, and evaluate its contrast enhancement with the microvessel confinement.

5. Conclusion

We report the investigation of PCCA-produced ultrasound contrast enhancement after acoustic activation with and without a microvessel confinement on an experiment phantom. The results demonstrate that the acoustic vaporisation of PCCAs in the 200 μm cellulose tube was very different from that in the open environment, with contrast enhancement in PI signals suppressed by 19.5-fold. The contrast enhancement in fundamental B-mode shows an even larger difference (27.4-fold). This investigation and quantification provide improved understanding of vaporising and imaging PCCAs in a more realistic microenvironment.

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ORCID iDs

Shengtao Lin  <https://orcid.org/0000-0003-2194-6965>

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